NYU

Music Recommendation System

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12/10/2024

Introduction

- Purpose: Help user to discover new music that they will like
- Objectives: Build a system combining personalization (content-based) and trend detection (collaborative filtering)

Data Source

- Data Source: Spotify Million Playlist Dataset & Spotify API
- Include playlist_id, playlist_name, track_id, album_name, etc.
- Pick 4000 out of 1000,000 playlists whose number of songs are less than 40 songs and create **playlists.csv**
- Use **track_id** for each song in the playlist to retrieve more information (**audio features** such as ‘Acousticness', 'Instrumentalness', 'Liveness', 'Valence', 'Tempo', etc.) through **Spotify API** and create **tracks.csv**

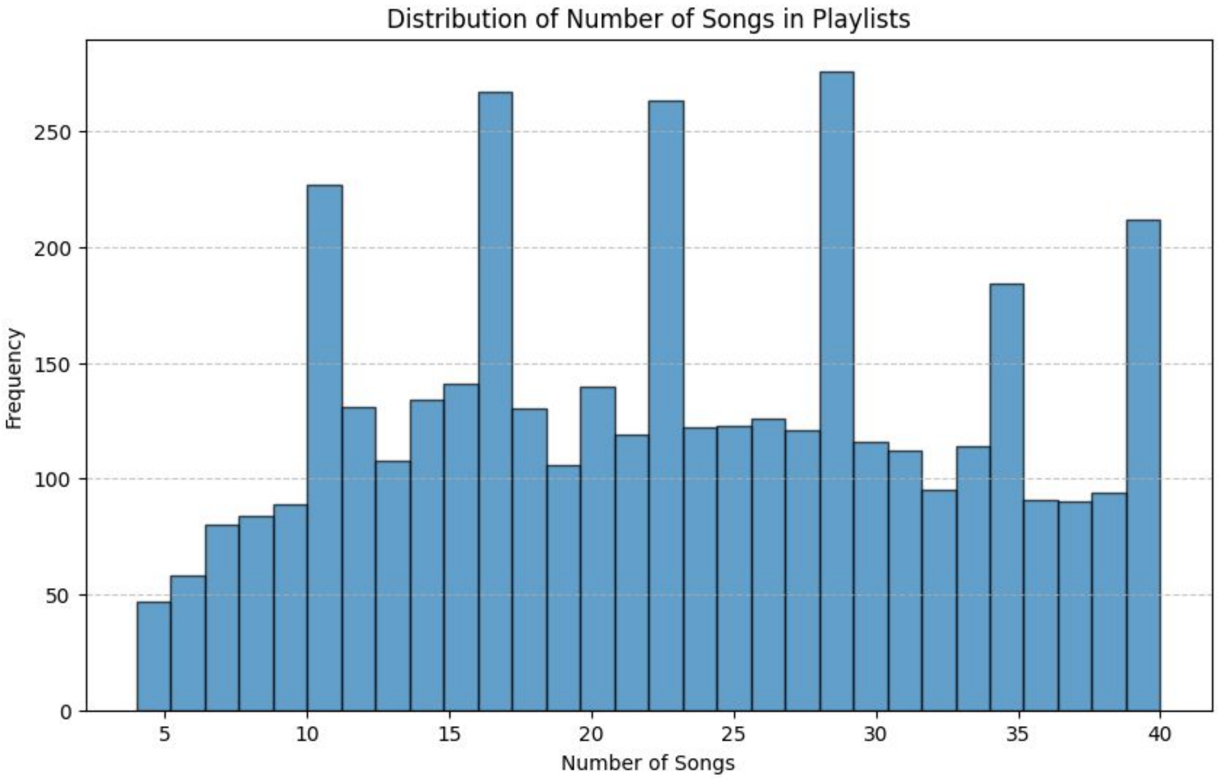
Data Source


```
{
        "name": "musical",
        "collaborative": "false",
        "pid": 5,
        "modified_at": 1493424000,
        "num_albums": 7,
        "num_tracks": 12,
        "num_followers": 1,
        "num_edits": 2,
        "duration_ms": 2657366,
        "num_artists": 6,
        "tracks": [
            {
                "pos": 0,
                "artist_name": "Degiheugi",
                "track_uri": "spotify:track:7vqa3sDmtEaVJ2gcvxtRID'
                "artist_uri": "spotify:artist:3V2paBXEoZIAhfZRJmo2
                "track_name": "Finalement",
                "album_uri": "spotify:album:2KrRMJ9z7Xjoz1Az406UML'
                "duration_ms": 166264,
                "album_name": "Dancing Chords and Fireflies"
            },
```

Data Source

	Track_ID	Popularity	Release Date	Explicit	Danceability	Energy	Key	Loudness	Mode	Speechiness	Acousticness	Instrumentalness	Liveness	Valence	Tempo
0	44uuZDQFAtfag94mDPIsEu	0	2017-06-14	TRUE	0.571	0.578	8	-9.696	0	0.27	0.254	1.78E-06	0.113	0.338	133.731
1	0Vwfd6fxFrL3kCnZSJ9vid	0	2013-01-01	TRUE	0.74	0.716	6	-5.796	0	0.0609	0.207	3.51E-06	0.0926	0.804	117.977
2	2Ze0YvSXz8CnC81hw5rXNo	17	2015-10-16	FALSE	0.61	0.766	2	-5.663	1	0.0267	0.0233	0.0148	0.107	0.582	107.483
3	34hMOtKwf5nm8tjvkGV0Dk	46	2016-05-06	FALSE	0.706	0.324	6	-14.048	0	0.0305	0.167	0.104	0.116	0.114	119.992
4	6EpRaXYhGOB3fj4V2uDkMJ	0	2017-05-18	FALSE	0.869	0.485	6	-5.595	1	0.0545	0.246	0.0	0.0765	0.527	106.028
5	6lvsJDZ7336YmpBzcNGhbe	62	2016-03-25	FALSE	0.52	0.513	7	-7.208	1	0.0256	0.281	0.0	0.0976	0.17	77.903
6	41d2Q6DHcM20OdzynkRtvf	1	2016-09-16	TRUE	0.662	0.837	8	-5.015	1	0.0582	0.0623	0.0	0.0931	0.613	136.998
7	07Y1uIQqxFbvSKzqLzqJQb	21	2014-08-08	FALSE	0.574	0.547	0	-8.134	1	0.0435	0.607	2.09E-06	0.0762	0.236	118.14
8	44f2AVSNQ3xVs1EI0S0bjl	0	2014-07-20	FALSE	0.569	0.183	7	-14.371	1	0.0298	0.914	0.404	0.106	0.13	70.06
9	4cxvludVmQxryrnx1m9FqL	67	2015-09-25	TRUE	0.698	0.649	8	-6.764	1	0.415	0.15	0.0	0.0903	0.568	180.466

Data Source



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Data Pre-Processing

- Use **MinMaxScaler** to transform the numeric audio features in the tracks dataset
- Playlists were split into train and test sets by 80% and 20%.
- The recommendation system is trained on the playlists in the train set and recommend tracks that were not in the train set. We evaluate the model by comparing the results to the test set

Approach

Content-Based Filtering - Focuses on track features to find tracks with similar characteristics. Recommend tracks similar to those the user already likes

- Build User Profile
  - Aggregate the features of tracks in the user's playlist to represent their preferences.
- Compute Similarity
  - Use cosine similarity to compare the user profile with candidate tracks.

Approach

Collaborative Filtering - Focuses on relationships between tracks based on their co-occurrence in playlists. Tracks frequently appearing together are likely to be recommended.

- Co-Occurrence Analysis:
  - Build a matrix showing how often pairs of tracks appear in the same playlist.
- Score Computation:
  - Assign higher scores to tracks that co-occur more often with the user's playlist tracks.

Approach

Hybrid Method

- Combination of content-based and collaborative filtering
- Recommends both familiar (content-based) and new (collaborative) tracks

Hybrid Score = α * Content Score + (1 - α) * Collaborative Score

α is the weight parameter to control the balance

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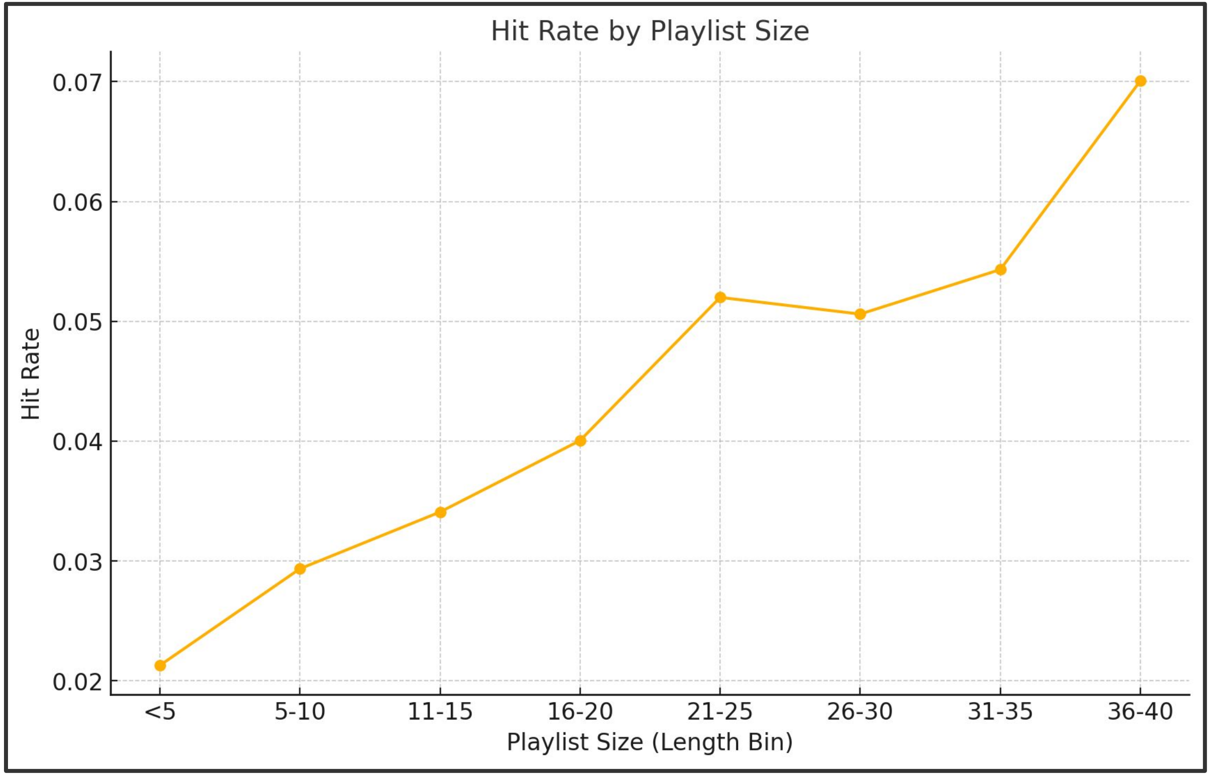
Results and Evaluation

- For each test case, the system recommended the same number of tracks as those withheld in the test set, ensuring a fair comparison between predicted and actual preferences.
- The best α to control the balance of collaborative filtering and content-based filtering is **0.60**
- The system's recommendation accuracy shows positive correlation with playlist size, suggesting the model performs better for users with larger playlists

Results and Evaluation



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Results and Evaluation

Recommended Songs

	Artist	Track Name	Album
1	Future	Mask Off	FUTURE
2	Kendrick Lamar	HUMBLE.	DAMN.
3	Post Malone	Congratulations	Stoney
4	Migos	Bad and Boujee (feat. Lil Uzi Vert)	Culture

Test Tracks

	Artist	Track Name	Album
1	Big Sean	I Don't Fuck With You	Dark Sky Paradise
2	Metro Boomin	No Complaints	No Complaints
3	Post Malone	Congratulations	Stoney
4	Post Malone	Go Flex	Stoney

Conclusion

- Hybrid approach outperforms both collaborative and content-based method alone, indicating the value of combining multiple strategies for more accurate music suggestions
- Content-based systems rely on extracting audio features (e.g., tempo, timbre) or metadata (e.g., genre, artist), which may not fully capture what makes a song appealing. Complex aspects like lyrics meaning, cultural trends, or emotional impact are difficult to capture with content-based features alone
- Possible improvements could result from a more complex model such as using NLP for lyrics analysis to extract more features or getting more user's explicit feedback, such as like or dislike and ratings